

Feature Dictionary: Processed Datasets

This dictionary defines the features available in the optimized Parquet artifacts located in `data/processed/`. These files have been cleansed, imputed, and optimized for downstream modeling (RecSys, RFM, and Clustering).

1. Interaction Data (`train_processed.parquet` / `test_processed.parquet`)

- Purpose: Transactional logs with interaction strength and temporal weighting.
- Split: `train` (interactions before 2020-08-25), `test` (interactions from 2020-08-25 to 2020-09-22).

Feature	Type	Null %	Range / Uniques	Sample	Purpose
<code>t_dat</code>	Date	0.00%	734 unique days	2018-09-20	Transaction Date
<code>customer_id</code>	Int32	0.00%	[0, 1,371,979]	512	Mapped Customer ID
<code>article_id</code>	Int32	0.00%	[0, 105,541]	2048	Mapped Article ID
<code>price</code>	Float64	0.00%	[0.00001, 0.591]	0.0508	Normalized Price
<code>sales_channel_id</code>	Int8	0.00%	[1, 2]	2	Channel (1: Online, 2: Store)
<code>quantity</code>	UInt32	0.00%	[1, 570]	1	Number of units in transaction
<code>days_ago</code>	Int64	0.00%	[0, 733]	733	Days since global max date
<code>time_weight</code>	Float64	0.00%	[0.0, 1.0]	0.00000004	Exp decay weight (30d half-life)

2. User Metadata (`customers_processed.parquet`)

- Purpose: Cleansed and imputed customer profiles.

Feature	Type	Null %	Range / Uniques	Sample	Purpose
<code>customer_id</code>	Int32	0.00%	[0, 1,371,979]	512	Mapped Primary Key

Feature	Type	Null %	Range Uniques	/ Sample	Purpose
FN	Int8	0.00%	[0, 1]	1	Fashion News flag (Imputed Null → 0)
Active	Int8	0.00%	[0, 1]	1	Active profile flag (Imputed Null → 0)
club_member_status	String	0.44%	4 unique	ACTIVE	Membership status
fashion_news_frequency	String	1.17%	5 unique	NONE	News frequency
age	Int8	0.00%	[16, 99]	49	Customer age (Affinitive Imputation)
postal_code	String	0.00%	352,900 unique	Unknown	Hashed location (Masked Super-Code)

3. Product Metadata (**articles_processed.parquet**)

- Purpose: Master catalog with mapped IDs.

Feature	Type	Null %	Range Uniques	/ Sample	Purpose
article_id	Int32	0.00%	[0, 105,541]	2048	Mapped Primary Key
product_code	Int64	0.00%	[108775, 959461]	108775	High-level SKU identifier
prod_name	String	0.00%	45,875 unique	Strap top	Product Name
product_group_name	String	0.00%	19 unique	Garment Upper body	High-level category
colour_group_name	String	0.00%	50 unique	Black	Primary color

Feature	Type	Null %	Range / Uniques	Sample	Purpose
index_name	String	0.00%	10 unique	Ladieswear	Macro department
section_name	String	0.00%	56 unique	Womens Everyday Basics	Store section
garment_group_name	String	0.00%	21 unique	Jersey Basic	Manufacturing group
detail_desc	String	0.39%	43,405 unique	Jersey top...	Product description